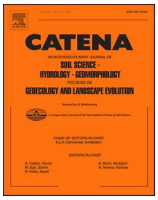
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Global soil erodibility factor (*K*) mapping and algorithm applicability analysis

Miaomiao Yang [a,b], Qinke Yang [a,b,*], Keli Zhang [c], Guowei Pang [a,b], Chenlu Huang [d]

[a] College of Urban and Environmental Science, Northwest University, Xi'an, 710127, China
[b] Key Laboratory of State Forestry Administration on Ecological Hydrology and Disaster Prevention in Arid Areas, Xi'an, 710127, China
[c] College of Geography, Beijing Normal University, Beijing, 100875, China
[d] College of Tourist (Institute of Human Geography), Xi'an International Studies University, Xi'an, 710127, China

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ABSTRACT

The soil erodibility factor (*K*) is the main data required for regional soil erosion investigation and mapping using soil erosion models. Fine mapping of *K* and the study of the applicability of different *K* estimation methods at the global scale are important to improve the accuracy of global soil erosion evaluation. In this study, the USLE-K, RUSLE2-K, EPIC-K and Dg-K algorithms were used to calculate and compare the global *K*, and a global measured *K* database was established by using literature backtracking method and retrieval tool method. Spatial pattern and applicability analysis were carried out on the results of the above four algorithms. The four algorithms were corrected according to the measured *K* database. The results showed that (1) the global *K* spatial patterns obtained by the four algorithms were similar but slightly different, with the result of RUSLE2-K being the closest to the measured *K*, followed by the USLE-K and the EPIC-K, and the result of Dg-K differing significantly from the measured *K*. (2) The global *K* distribution characteristics showed some regularity with soil properties, such as soil silt content and sand content, with silt content having the greatest influence on *K*. (3) The results calculated by the corrected RUSLE2-K and USLE-K algorithms could meet the model applicability conditions and coincided with the results of local *K* mapping. The results of *K* mapping in this study on a global scale and the results of the comparative analysis of the applicability of different algorithms provide the necessary scientific basis for the selection of *K* algorithms globally and quantitative evaluation of soil erosion.

1. Introduction

Soil erosion is the most serious land degradation and environmental problem in the world today (FAO, 2019; Pimentel and Burgess, 2013), and approximately three quarters of global land degradation is related to soil erosion (Bruce et al., 1995; IPBES, 2018). The control of soil erosion should be based on soil erosion evaluation and mapping (Borrelli et al., 2017; Oldeman et al., 1994), and the calculation and analysis of soil erodibility is the basis of soil erosion mapping. Soil erodibility can be measured by a variety of indexs (Morgan, 2005), such as dispersion ratio, instability index, soil cohesion, soil erodibility factor, etc., among which, soil erodibility factor (*K*) introduced by Universal Soil Loss Equation (USLE) can be directly used in soil erosion prediction models. *K* is one of the most essential parameters of the model when applying soil erosion prediction models for soil erosion assessment and mapping (Liu et al., 2020; Oldeman et al., 1994; Panagos et al., 2014; Römkens et al., 1997; Yang et al., 2020).

K is a comprehensive parameter that represents an integrated average annual value of the total soil and soil profile reaction to a large number of erosion and hydrologic processes (Römkens et al., 1997) and is a key parameter in determining soil loss rates (Römkens et al., 1997; Wischmeier et al., 1971; Wischmeier and Smith, 1978). *K* can be obtained through a variety of techniques, including calculation based on measurement of soil physicochemical attributes, measurement of scourability, measurement by rainfall simulation, and field plot tests (Zhang et al., 2007). Currently, long-term monitoring of soil loss data is mainly obtained through data from field runoff plots under natural rainfall conditions. Direct measurement of *K* requires long-term erosion observation in field plots and is costly and time-consuming. Therefore, to quantitatively assess soil erosion at large regional scales, scholars have conducted a series of *K* mapping studies based on soil property datasets (Zhang et al., 2008). Global studies include global *K* mapping

* Corresponding author.
E-mail address: qkyang@nwu.edu.cn (Q. Yang).

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Table 1
List of basic data.

Data	Source	Resolution	Use
Clay content (%)	Download from ISRIC website(1)	7.5 arc seconds	For calculating the K
Silt content (%)	As above	As above	As above
Sand content (%)	As above	As above	As above
Organic matter content (g/kg)	As above	As above	As above
Texture class	As above	As above	As above
Soil type (WRB)	As above	As above	Comparison of K in different soil types
Soil structure class	The results of the project team's work	As above	Calculation of K for the USLE-K, RUSLE2-K algorithms
Rainfall and runoff factors (MJ. mm)/(ha.h.yr)	As above	As above	For extrapolating the measured K
Measured K (t.hm^2. h)/(hm^2.MJ. mm)	Excerpts from the literature	––	Verify the applicability of K algorithm

Note: Soil data can be downloaded at:(1) ISRIC–International Soil Reference and Information Centre: https://files.isric.org/soilgrids/latest/.

by Nachtergaele et al. (2011), national and large regional mapping studies such as China *K* mapping by Liang et al. (2013); EU *K* mapping by Panagos et al. (2014); Eastern China *K* mapping by Wang et al. (2016); Australian *K* mapping by Yang et al. (2018); Greek *K* mapping by Efthimiou (2020); Brazilian *K* mapping by Godoi et al. (2021); and Southeastern China *K* mapping by Zhu et al. (2021). These mapping studies have provided strong support for regional *K* mapping using methods such as *K* calculation and spatial interpolation (or spatial prediction by machine learning methods) based on point soil property data and map algebra-based *K* calculation using surface soil property datasets.

The Universal Soil Loss Equation (USLE) is one of the most commonly used predictive models for soil erosion to date (Alewell et al., 2019; Borrelli et al., 2021; Lal, 1994), and the *K* algorithm presented in the USLE (USLE-K) is one of the most commonly used algorithms for calculating *K*. Renard and Dabney (2013) proposed an updated version of the *K* algorithm (RUSLE2-K) in Revised Universal soil Loss Equation version 2 (RUSLE2), which basically follows the USLE-K algorithm and improves the computation of the soil texture component, soil profile permeability component, and so on. In order to determine the relationship between soil erosion and soil productivity, Sharpley and Williams (1989) developed the Erosion Productivity Impact Calculator (EPIC), in which the EPIC-K algorithm estimates *K* only using soil organic carbon and particle size property. In order to further simplify the calculation of *K*, Römkens et al. (1997) used the available erodibility factor database obtained from natural runoff plots and simulated rainfall studies, and proposed the Dg-K algorithm based on geometric mean diameter (Dg)-based model that only considered the particle size property in the analysis, which could be used to calculate *K* in the case of insufficient basic data.

However, since most of the algorithms for calculating *K* (e.g., USLE-K (Wischmeier and Smith, 1978) and RUSLE2-K (Renard et al., 1997)) were developed on the basis of soil conditions in the United States, whether these algorithms are applicable to other countries, regions, or global scales and whether they need to be corrected based on existing algorithms have received scholarly attention (Borselli et al., 2012; Sanchis et al., 2010; Zhang et al., 2016; Ghosal et al., 2020; Corral-Pazos-de-Provens et al., 2023; Tian et al., 2023). Related studies compared the estimated *K* using algorithms such as USLE-K, RUSLE-K, EPIC-K, and Dg-K with the measured *K* of natural runoff plots, etc., and found that the estimated *K* were larger than the measured *K*. For example, Wang et al., (2013),Vaezi et al. (2008), Hussein et al. (2007) and Rejman et al. (1998) found that the USLE-K algorithm overestimated soil *K* in China, northwestern Iran, northern Iraq and southeastern Poland, respectively. Bagarello et al. (2012) concluded that the measured K value for the clay soil at the Sparacia (Sicily) was 1.85 times higher than the *K* obtained by the USLE-K algorithm. Yang et al. (2018) found that the RUSLE-K algorithm gave better estimation results in estimating *K* in Australia. Some studies have also analyzed the applicability of different *K* algorithms using *K* data from relevant databases, e.g., Torri et al (1997) and Auerswald et al. (2014) investigated the accuracy of the USLE-K algorithm for *K* prediction using a global *K* dataset extracted from the literature worldwide and a soil dataset from Central Europe and found that its accuracy was low. Kinnell (2016) used event data from paired bare fallow plots in the USLE database to examine the abilities of replicate plots, the USLE, and the USLE-M to model event erosion on bare fallow plots, and found that the replicate model performed better than the USLE-M. Raj et al. (2023) compared the K values with the available literature and found that the *K* obtained from USLE-K was more applicable to Indian compared to EPIC-K. The above results indicate that the *K* methods and equations (especially USLE-K, RUSLE-K and EPIC-K) based on the U.S. runoff plots and artificial rainfall test database may have some problems when applied outside the U.S., and the estimated *K* needs to be corrected based on the field-measured data, if the conditions permit. However, the necessary research is lacking in the comparison and applicability analysis of various *K* algorithms on a global scale, as well as high-resolution mapping on a global scale and its comparative analysis with local mapping.

Considering that existing *K* estimation models have been proposed and widely applied, high-resolution soil property data covering the world have also been available and updated, and a number of measured *K* data have been accumulated. It is possible for us to map *K* on a global scale and perform algorithm adaptability analysis based on a variety of algorithms. This study uses the latest (2020) global soil property datasets published by the International Soil Reference and Information Centre (ISRIC) to calculate the global *K* using the algorithms of the USLE, RUSLE2, EPIC, and Dg models on a global scale. The calculation of the global *K* is carried out, and the results are analyzed for their reasonableness and applicability using measured data reported in the literature. This study will support the rational application of the algorithm in *K* mapping for global and regional soil erosion mapping and thus improve the accuracy of global and regional soil erosion mapping.

2. Methodology

2.1. Study area and data sources

This study was carried out worldwide but did not include the Antarctic continent and Greenland, which are covered with ice and snow year round, have small permanent populations and feature soil erosion that is still in a natural stage of erosion. The basic datasets used in this study included the soil properties (Table 1). Soil property datasets are the latest (2020) global data SoilGrids 2.0 (Poggio et al., 2021) and partial SoilGrids 1.0 (Hengl et al., 2017) released by the International Soil Reference and Information Centre (ISRIC), which were obtained by machine learning based on 240,000 soil profiles collected worldwide under a unified standard. Compared to previous similar datasets, such as the 1:5 million world soil map (FAO-Unesco, 1974), the Global Soil Database (5 km resolution, Harmonized World Soil Database, FAO, 2012)and soil census data from various countries (China, USA, etc.) (USDA-NRCS, 2019; Zhu, 1996), SoilGrids data are of high resolution, have more complete attributes and are more compatible with the main soil classification systems (such as US Taxonomy and World Reference Base (WRB) for Soil Resources by FAO) and have been updated to be more suitable for this study. The measured datasets extracted from the literature are shown in Appendix Table A.1.

Table 2
Upper and lower limits and ranges for each particle size class.

Particle size class	Particle size minimum d_i/mm	Particle size maximum d_{i-1}/mm	Mean particle size m_i/mm
Clay	0.00005	0.002	0.001
Silt	0.002	0.05	0.026
Sand	0.05	2	1.025

Table 3
Table of permeability class-texture class relationship (USDA-NRCS, 1983).

Number of texture class	Texture class	Permeability class	Number of texture class	Texture class	Permeability class
1	Clay	6	7	Loam	3
2	Silt Clay	6	8	Silt Loam	3
3	Sand Clay	5	9	Sand Loam	2
4	Clay Loam	4	10	Silt	1
5	Silt Clay Loam	5	11	Loam Sand	2
6	Sand Clay Loam	4	12	Sand	1

2.2. Calculation of the soil erodibility factor (K)

2.2.1. Algorithm of the soil erodibility factor (K)

Four algorithms, which are relatively mature and widely used in previous studies (Wang et al., 2013), were used in this study. They are briefly introduced as follows.

(1) USLE-K algorithm.

Wischmeier et al. (1971) proposed a nomogram for calculating K using soil mechanical composition, soil organic matter content, soil structure and permeability characteristics and fitted it to a model for calculating USLE-K (Wischmeier and Smith, 1978).

$$K = \frac{2.1 \times 10^{-4} M^{1.14}(12 - OM) + 3.25(s-2) + 2.5(p-3)}{100} \times 0.1317 \quad (1)$$

$$M = \left(M_{silt} + M_{vfs}\right)*(100 - M_c) \quad (2)$$

where K is the soil erodibility factor (t.hm^2.h)/(hm^2.MJ.mm), OM – soil organic matter content (%) = soil organic carbon (‰) * 0.1724; p – permeability class; s – soil structure class; M_{vfs} –very fine sand content, very fine sand grain content = sand content * 0.2; M_{silt} – silt content (%); M_c – clay content (%).

(2) RUSLE2-K algorithm.

The RUSLE2-K algorithm was proposed by the RUSLE2 manual (Renard et al., 1997) and is a modified version of the USLE-K algorithm. This algorithm adjusts the soil structure classification k_s and soil texture component k_t of the USLE-K algorithm, and its application scope is consistent with USLE-K, as follows:

$$K = 0.1317 \times \left(K_t K_o + K_s + K_p\right)/100 \quad (3)$$

where K is the soil erodibility factor; k_t is the texture fraction; k_0 is the organic matter fraction; k_s is the soil structural fraction; and k_p is the soil permeability fraction.

Soil texture fraction: The soil texture fraction is calculated using the following equation (Renard and Dabney, 2013):

$$k_t = \begin{cases} k_{tb} & M_{silt} + M_{vfs} \le 68\% \\ k_{tb} - 0.67\left[\left(k_{tb} - k_{t68}\right)^{0.82}\right] & M_{silt} + M_{vfs} > 68\% \end{cases} \quad (4)$$

$$K_{tb} = 2.1 \times 10^{-4}\left[\left(K_{silt} + M_{vfs}\right) \times \left(100 - M_{elay}\right)^{1.14}\right] \quad (5)$$

$$K_{t68} = 2.1 \times 10^{-4}\left[\left(68 \times \left(100 - M_e\right)\right)^{1.14}\right] \quad (6)$$

$$M_{vfs} = \left(0.74 - 0.62 \times M_{sand}/100\right) \times M_{sand} \quad (7)$$

where M_{silt} is the silt content (%); M_{vfs} is the very fine sand content (%); M_{clay} is the clay content (%); and M_{sand} is the sand content (%).

Soil organic matter fraction: The soil organic matter fraction is calculated using the same algorithm as the USLE-K model:

$$k_o = 12 - OM \quad (8)$$

where OM is the soil organic matter content (%).

Soil structural fraction: The soil structural fraction is calculated using the following equation:

$$K_s = 3.25(2 - s) \quad (9)$$

where s is the soil structure class.

Soil profile permeability fraction: The soil permeability factor can be calculated using the following equation.

$$K_p = 2.5(p - 3) \quad (10)$$

where p is the permeability class.

(3) EPIC-K algorithm.

The EPIC-K model proposed by Sharpley and Williams (1989), using which only data related to soil organic carbon content and particle composition information are required to complete the estimation of K, is calculated as follows:

$$K = 0.1317 \times \{0.2 + 0.3 \times \exp[-0.0256 \times SAN \times (1 - SIL/100)]\} \times \left(\frac{SIL}{CLA + SIL}\right)^{0.3} \times \left[1.0 - \frac{0.25 \times C}{C + \exp(3.72 - 2.95 \times C)}\right] \times \left[1.0 - \frac{0.7 \times SN1}{SN1 + \exp(-5.51 + 22.9 \times SN1)}\right] \quad (11)$$

where CLA is the clay content (%)、SIL is the silt content (%)、SAN is the sand content (%) ; $SN1 = 1 - SAN/100$; C is the soil organic carbon content (%).

(4) Dg-K algorithm.

The algorithm was proposed by Römkens et al. (1997) in the RUSLE manual with the following expression:

$$Dg = \exp\left(0.01 \times \sum_{i=1}^{n}(fi \times \ln(mi))\right) \quad (12)$$

where D_g is the geometric mean particle diameter (mm), m_i (mm) is the average of the maximum and minimum range of values for a given particle size (Table 2), f_i is the mass percentage (%) of a particle size, and n is the total number of particle sizes (three in total, clay, silt and sand).

The global algorithm based on the geometric particle size mean K proposed by Römkens et al. (1997), fitted by 225 groups of soils on a global scale, is expressed as follows:

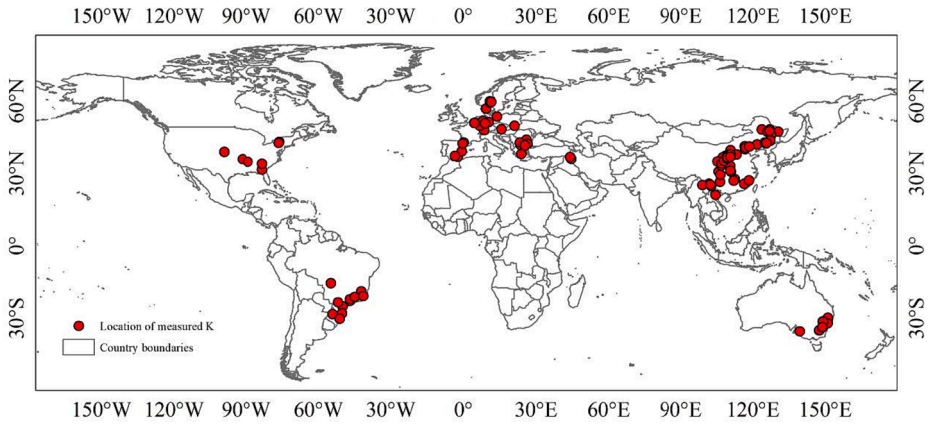


Fig. 1. Global distribution of measured *K* sampling points.

$$K_{Dg} = 0.1317 \times 7.594 \times \left\{ 0.0034 + 0.0405 \times exp\left[-\frac{1}{2} \times \left(\frac{\log(Dg) + 1.659}{0.7101} \right)^2 \right] \right\} \quad (13)$$

where log is the logarithm with a base of 10; the calculation results in U.S. units, after the completion of the calculation to give the results * 0.1317 converted to the international system ((t.hm^2.h)/(hm^2.MJ.mm)).

2.2.2. Data preprocessing

Before the calculation, the necessary preprocessing of the data needs to be performed, and all these processing steps are performed in K_Tool developed by the author (Yang and Zhang, 2019), including the following:

soil structure class (*s*): Based on the soil profile records in the manual of the World Soil Map (FAO-Unesco, 1974), the relationship table between the main soil types and s values in the WRB soil systematic classification was sorted out. Then, using the pedotransfer function (McBratney et al., 2002; Van, 2017), the map of s corresponding to the WRB soil type maps was generated.

Silt content: One of the constraints of the USLE-K algorithm is a silt content < 70 %; when the silt content > 70 %, it is directly assigned to 70 %.

Extrapolation of very fine sand content (Panagos et al., 2014**):** very fine sand content = sand content * 0.2.

Soil organic matter content (Wischmeier et al., 1971**):** cannot be greater than 4 %; when the soil organic matter content is greater than 4 %, it is assigned directly to 4 %.

Permeability class: This class is determined according to the soil texture class and corresponds to Table 3.

2.2.3. Calculation of K

In this study, the calculation tools based on ArcToolBox and C language (K_Tool and Dg-Box) developed by the project team in the early stage (Yang and Zhang, 2019) were used to calculate *K* of different algorithms. The parameters required for the USLE-K and RUSLE2-K algorithms include: clay content (%), silt content (%), sand content (%), organic carbon content (g/kg), soil texture class and soil structure class. EPIC-K algorithm only needs to input clay content (%), silt content (%), sand content (%) and organic carbon content (g/kg), which are all input in raster layer. The raster layer of *K* values of USLE-K, RUSLE2-K and EPIC-K algorithms can be obtained after calculation. For the Dg-K algorithm, the Dg-Box tool only needs to input the raster data of clay content (%), silt content (%) and sand content (%) to obtain the *K* raster layer.

2.3. Establishment of the measured K database

(1) Literature search and extraction

Using search platforms such as the Web of Science and China Knowledge Network, we used the key words soil loss, soil erodibility, and runoff plot and combined them with the literature backtracking method (literature cited in important literature). We reviewed 340 Chinese and English studies on runoff plot observations and soil erosion surveys in small watersheds (Table A.1). The 340 results reviewed were screened, and only the actual measured *K* obtained in runoff plots under natural or simulated rainfall conditions were retained. The literature and data with *K* calculated according to four algorithms, USLE-K, RUSLE2-K, EPIC-K and Dg-K, were excluded. Some of the literature contains only measured data on the soil erosion rate and does not contain measured *K*, which needs to be calculated based on the soil erosion rate. In this case, literature that does not record slope, slope length, or land use type needs to be excluded. After the screening of the above two conditions, 144 records were eligible for the literature, with a total of 1260 runoff plots for a total of 20,936 plot years, and the distribution of sample points is shown in Fig. 1. Soil erosion observation (plot) data from all over the world, including geographical location (latitude and longitude, country, city), runoff plot number, climate zone (Köppen climate type zone), average annual precipitation, land use type, biological/engineering measures, slope length (m), slope (in degrees or %), runoff plot area (m^2), measured water erosion loss rate or *K*, starting year of observation, ending year of observation, and data source, were extracted from the screened literature, and a global database of measured *K* was initially built.

(2)Data collection

Data collected for each influence factor in the global measured *K* database need to be extrapolated, including the following:

① Soil water erosion rate (*A*): multiyear average soil loss per unit area, with a scale of t/(ha.yr), in agreement with the USLE (Wischmeier and Smith, 1978) and RUSLE (Renard et al., 1997). The *A* values extracted in the literature have different scales and were finally converted to t/ha.a uniformly.

② Rainfall and runoff factor (*R*): In the USLE (Wischmeier and Smith, 1978) series of models, the *R* factor is the value of the rainfall erosivity index plus a factor for any significant runoff from snowmelt and is influenced by the combination of rainfall intensity, rainfall volume, and snowmelt (ice) runoff. The daily, monthly or annual rainfall erosion force was calculated based on daily rainfall data from meteorological stations, and then the continuous *R* surface was obtained by the spatial interpolation method with units of (MJ.mm)/(ha.h.yr). In this study, the method of Yin et al. (2019) provided by the project team was used to calculate the global *R*, which was sampled for each point after the necessary preprocessing (null elimination, low-pass filtering, etc.).

③ Slope length and slope factor (*L*, *S*): According to the USLE

Table 4
C values for nonagricultural land.

IGBP Land-cover type	MIN_c-MAX_c	IGBP Land-cover type	MIN_c-MAX_c
Water	0	Open shrublands	0.01 ~ 0.15
Evergreen Needleleaf forest	0.0001 ~ 0.003	Woody savannas	
Evergreen Broadleaf forest		Savannas	0.01 ~ 0.15
Deciduous Needleleaf forest		Grasslands	
Deciduous Broadleaf forest		Permanent wetlands	0
Mixed forest		Urban and built-up	
Closed shrublands	0.01 ~ 0.15	Snow and ice	

Table 5
C factor assignment table (According to Borrelli et al., 2017).

Crop Group			C-factor
		Various	0.2
		Maize	0.38
		Rice	0.15
2	Legume Vegetables	Various	0.32
3	Root and Tuber Vegetables	Various	0.34
4	Fruiting Vegetables	Various	0.25
5	Cucurbit Vegetables	Various	0.25
6	Bulb Vegetable	Various	0.3
7	Leafy Vegetables	Various	0.25
		Tobacco	0.5
8	Forage, Fodder and Straw of Cereal Grains Group	Mixed-legumes	0.15
		Mixed-grasses	0.1
9	Grapes and Hops	Grapes	0.35
		Hops	0.42
10	Oilseed Group	Various	0.25
		Cotton	0.4
11	Fibre Crops	Fibre Crops	0.28
12	Berries Group	Various	0.15
		Strawberries	0.2
13	Shrubs Herbs and Spices Group	Shrubs Herbs and Spices	0.15
		Coffee	0.2
14	Trees/Fruit Tree	Various	0.15

Table 6
P factor assignment (Wischmeier and Smith, 1978).

Slope(%)	Contouring	Strip cropping	Terracing
1–2	0.60	0.30	0.12
3–8	0.50	0.25	0.10
9–12	0.60	0.30	0.12
13–16	0.70	0.35	0.14
17–20	0.80	0.40	0.16
21–25	0.90	0.45	0.18

Table 7
P factor assignments for selected measures (Xiong et al., 2019).

	paddy fields	orchards	croplands
P-factor	0.16	0.47	0.49

(Wischmeier and Smith, 1978) and RUSLE (Renard et al., 1997) manuals, the slope length factor (*L*) is the ratio of soil loss on the slope of a given length of field to loss on a 72.6-foot-long slope under the same conditions. The slope factor (*S*) is the ratio of soil loss on the slope of a field at a given slope to the loss on the slope at 9 % slope under otherwise identical conditions and expresses the potential influence of topography on soil erosion. In models such as USLE-K, RUSLE-K and CSLE-K, the basic meaning of *LS* factors is the same, but their algorithms differ slightly. Considering the relatively large variability of slope across the globe (slope variation (1.43° to 46.67°), mean (8.87°), median (5.14°) and dominant (5.14°) values in the measured *K* database) in the measured *K* database, the CSLE slope length factor algorithm specified in the Chinese water census (Center, 2020; Liu and Zhang, 2002) is sampled here and calculated point by point based on the slope and slope length from the literature data.

$$S = \begin{cases} 10.8\sin\theta + 0.03 & \theta < 5^{\circ} \\ 16.8\sin\theta - 0.5 & 5^{\circ} \le \theta < 10^{\circ} \\ 21.9\sin\theta - 0.96 & \theta \ge 10^{\circ} \end{cases} \quad (14)$$

$$L = (\lambda/22.1)^{m} \quad (15)$$

$$m = \begin{cases} 0.2 & \theta \le 1^{\circ} \\ 0.3 & 1^{\circ} < \theta \le 3^{\circ} \\ 0.4 & 3^{\circ} < \theta \le 5^{\circ} \\ 0.5 & \theta > 5^{\circ} \end{cases} \quad (16)$$

where ***S*** is the slope factor, θ is the slope value (°), *L* is the slope length factor, λ is the slope length (m), and *m* is the slope length index, which takes different values according to the slope.

④ Cover and management measures factor (*C*): *C* is the ratio of soil loss on a certain area of land under certain cover and management measures to the loss on the same area of land with continuous clear-cutting and recreational treatments (Renard et al., 1997; Wischmeier and Smith, 1978). The calculation of *C* should be specified or calculated based on the measured data. However, there are insufficient data on a global scale, so the method of Borrelli et al. (2017) was used for estimation as follows:

Nonagricultural land *C* factor (C_{na}): Refer to Borrelli et al. (2017) for the global crop cover factor and download the global vegetation continuous fields product (*VCF*, vegetation cover data) mean data on Google Earth Engine (GEE), which includes percent tree cover (*TC*, woodland cover surface) data and percent nontree vegetation (*NTC*, shrub, grass cover, etc.) data. After preprocessing the *TC* data and *NTC* data, each real measurement point is sampled, and *TC* data and *NTC* data are obtained based on the sampling.

For the calculation of *C* for nonagricultural land, the bounded range of *C* values corresponding to different land use types was determined by referring to the land use classification rules of the International Geosphere Biosphere Programme (IGBP) (Table 4). Based on the land use types extracted from the literature, forested land was calculated in Excel according to Equation (17), and both grassland and shrubland categories were calculated in Excel according to Equation (18).

$$C_{na} = MIN_c + (MAX_c - MIN_c)\cdot(1 - TC/100) \quad (17)$$

where C_{na} refers to the *C* value of the forest cover. The maximum MAX_c and minimum MIN_c values were set using the assignment rules proposed by Borrelli et al. (2017). The *TC* (woodland cover surface) was normalized to a range of 0 to 1.

$$C_p = MIN_c + (MAX_c - MIN_c)\cdot(1 - NTC/100) \quad (18)$$

where C_p refers to the *C* values for shrub and grass cover. The maximum MAX_c and minimum MIN_c values were set using the assignment rules proposed by Borrelli et al. (2017). *NTC* (shrub, grass cover surface) was normalized to a range of 0 to 1.

Agricultural land *C* factor (C_p): for agricultural land, according to the type of land use extracted from the literature, according to the table of *C* assignment for agricultural land in Borrelli et al. (2017) (Table 5), if no specific crop is given, the value is assigned to the mean value of *C* for agricultural land 0.26; if the specific crop is given in the literature, then the value is assigned separately. For example, for maize, the value will be 0.38.

⑤ Soil and water conservation measure factor (*P*): the ratio of soil loss under conditions where soil and water conservation measures

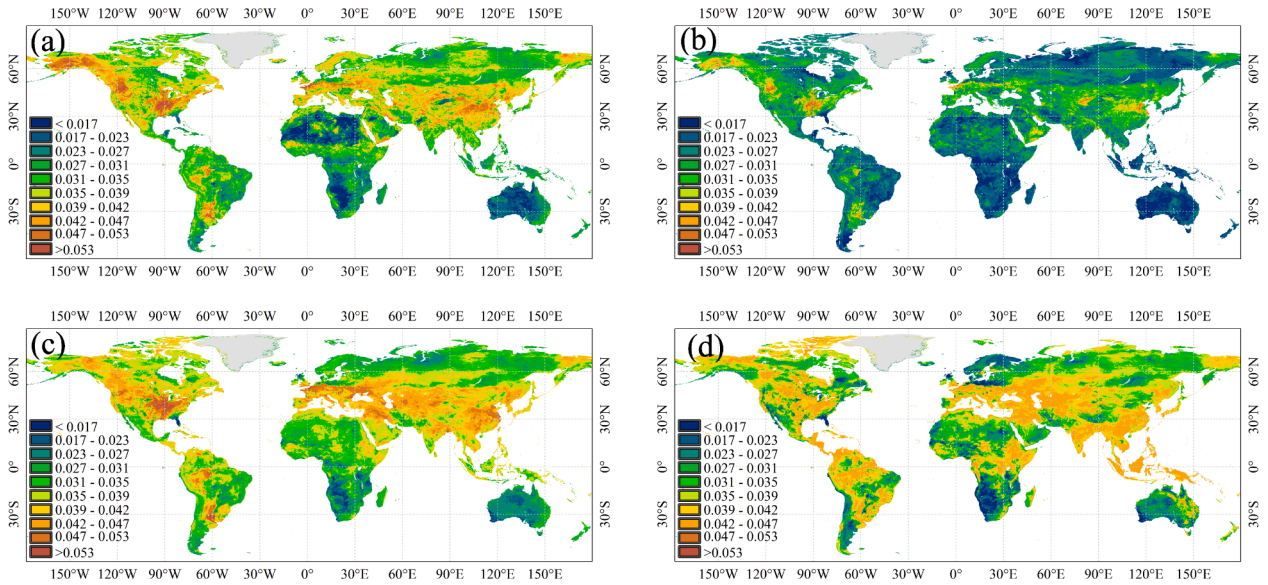


Fig. 2. Global *K* of different algorithm (a) USLE-K; (b) RUSLE2-K; (c) EPIC-K; (d) Dg-K.

such as contouring, strip cropping or terracing are adopted to the loss under downhill planting conditions (Renard et al., 1997; Wischmeier and Smith, 1978). Following the method of Xiong et al. (2019) and Wischmeier and Smith (1978), the *P* factor was assigned according to different types of engineering measures and slopes, as shown in Table 6.

Corresponding to terracing, contouring and strip cropping in the literature, various measures for different slopes were assigned according to Table 6; those without specific soil and water conservation measures were assigned according to the method of Xiong et al. (2019) (Table 7).

⑥ Soil erodibility factor (*K*): The soil erodibility factor (*K*) is a factor in the USLE/CSLE that refers to the rate of soil loss per unit rainfall erosion force for a given soil measured on a standard unit plot, the dimension of which is (t.hm^2.h)/(hm^2.MJ.mm) (Wischmeier and Smith, 1978).

$$A = R{\cdot}K{\cdot}L{\cdot}S{\cdot}C{\cdot}P \tag{19}$$

Under unit plot (slope length 72.6 feet, slope 9 %) conditions, $LSCP = 1$; thus,

$$K = A/R \tag{20}$$

However, the data we collected do not correspond exactly to standard plots, and we need to invert the *K* values by extrapolating the values of each factor through the process described above.

$$K = A/(R{\cdot}L{\cdot}S{\cdot}C{\cdot}P) \tag{21}$$

The above reduced *R*, *L*, *S*, *C* and *P* factors were substituted into Equation (21) to calculate the *K* values. The final results of 1260 runoff plots were obtained, corresponding to 144 measured sample locations and measured *K*. The soil types (WRB system) of the 144 measured sample locations mainly included Acrisols, Cambisols, Ferralsols and Luvisols.

2.4. Correction and accuracy evaluation of the K algorithm

(1) Correction of the *K* algorithm.

The applicability of the four algorithms was analyzed by comparing the estimated *K* values of USLE-K, RUSLE2-K, EPIC-K and Dg-K with the measured *K* values in terms of the range of values, the average value, and the difference with the measured *K* values; further fitting the estimated values of the four algorithms with the measured *K* to obtain the correlation equation between the estimated values and the measured *K*; correcting the estimated *K* values of USLE-K, RUSLE2-K, EPIC-K and Dg-K according to the measured *K*; and performing the evaluation of the model accuracy.

(2) Accuracy evaluation of the *K* algorithm.

In this study, the accuracy of the model is analyzed using the coefficient of determination (R^2) and the coefficient of model effectiveness (*NSE*) (Nash and Sutcliffe, 1970).

$$NSE = 1 - \frac{\sum_{i=1}^{n}(O_i - S_i)^2}{\sum_{i=1}^{n}(O_i - \overline{O})^2} \tag{22}$$

where O_i is the ith measured value; S_i is the corresponding ith model estimate; and $\tilde{O}$ is the mean of the observed values. In this study, we refer to Ahmad et al. (2011) and Wang et al. (2013), combine the characteristics of soil erosion research, and adopt $NSE > 0.4$ and $R^2 > 0.5$ as the criteria for the model to meet the requirements of forecast accuracy.

2.5. K Statistics of different soil types

The analysis of *K* of various soil types can provide support for rational utilization of soil resources and optimization of *K* algorithms. Based on ArcGIS 10.5 software, this paper uses spatial analysis tools to carry out average statistics on the *K* obtained by four algorithms of different soil types, to analyze soil types with higher and lower *K* and to analyze their reasons.

2.6. Analysis of the importance of soil properties for K

The random forest model is a classification tree-based machine learning algorithm proposed by Breiman (2001), which uses the bootstrap resampling method to draw multiple samples from the original sample, performs decision tree modeling on each bootstrap sample, and then combines the predictions of the multiple decision trees to arrive at the final prediction and the importance of the influential factors through voting. This study analyzes the importance of soil properties on *K* using the random forest model.

For the accuracy test of random forest regression, the evaluation metrics used in this study include the coefficient of determination (R^2) and the mean absolute error (*MAE*). The formulas are as follows:

$$\mathrm{R}^2 = 1 - \frac{\sum_{i=0}^{m}(y_i - \hat{y}_i)^2}{\sum_{i=0}^{m}(y_i - \overline{y})^2} \tag{23}$$

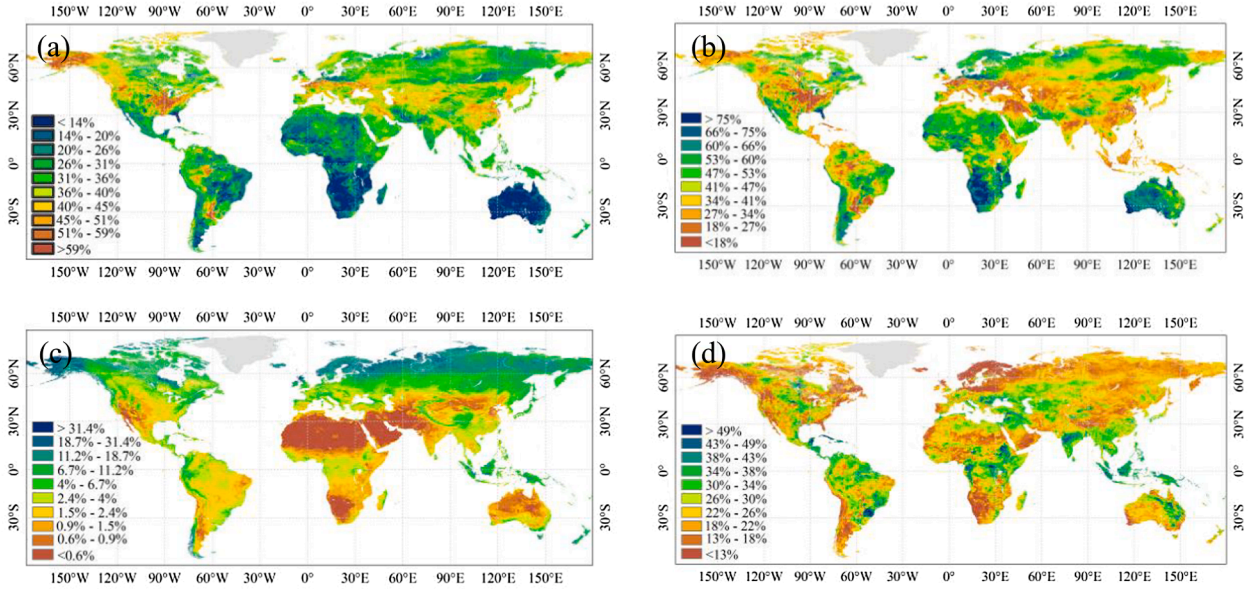


Fig. 3. Soil properties (a) Silt content; (b) Sand content; (c) Organic matter content; (d) Clay content.

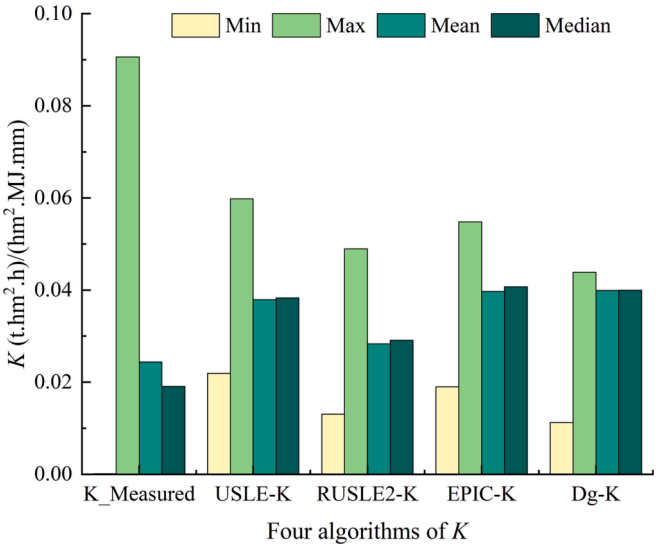


Fig. 4. Eigenvalues of measured K and the K of four algorithms.

$$MAE = \frac{1}{m}\sum_{i=0}^{m-1}|y_i - \hat{y}_i| \tag{24}$$

where y_i is the true value of the ith K; $\hat{y}_i$ is the prediction result of the ith K; $\bar{y}$ is the average value of K; and m is the total number of samples. The larger the value of R^2 is, the smaller the MAE is, and the higher the accuracy of the model explanation.

3. Results

3.1. Mapping results of four K algorithms

The global K map (Fig. 2) was obtained according to four different algorithms, USLE-K, RUSLE2-K, EPIC-K and Dg-K. First, the K values in the northern hemisphere are generally larger than those in the southern hemisphere, and the eastern part of South America, the southern part of Africa and the Australian continent all have low K values. Second, the spatial distribution of K obtained by the four algorithms is similar, showing a distribution pattern of lower K in the Brazilian highlands and Amazonian plains in South America, southern Africa and the Sahara desert, the Eastern European plains and Australia and higher K in the Canadian plains and Rocky Mountains in North America, the Western Siberian plains in western Europe, the Mongolian plateau in Asia, and northern China. The spatial pattern of K for the four algorithms also differs somewhat; for example, in southern Asia, the USLE-K and RUSLE2-K algorithms obtain relatively low K, and the Dg-K and EPIC-K algorithms obtain relatively high K. In the northern part of North America, northern Asia and central Africa, the Dg-K algorithm obtained relatively high K, and the other three algorithms obtained relatively low K. In central Asia, the USLE-K and RUSLE2-K algorithms obtained relatively low K, and the EPIC-K and Dg-K algorithms obtained relatively high K. This difference is the result of the subtle differences in the parameters of each model specified by the four algorithms.

A comparison with the spatial pattern distribution of soil properties (Fig. 3) revealed that the spatial pattern of K obtained by the USLE-K and RUSLE2-K algorithms was similar to that of the spatial distribution of silt content, with higher K in areas with higher silt content, such as central and southeastern Asia and central and western North America. The spatial pattern of K obtained by the EPIC-K and Dg-K algorithms is similar to that of the spatial distribution of sand content, with lower K values in areas with higher sand content, such as southern Europe, western Asia and central South America. The K obtained by the four algorithms is also relatively lower in areas with higher organic matter content and clay content, such as Australia, northern and southern Africa, and western and southern South America.

3.2. Measured results of K

The results of a total of 1260 runoff plots were collected in the global measured K database, including the calculated K according to the soil erosion factors and erosion rate and the measured K given directly in the literature, for a total of 144 measured sample sites worldwide. As shown in Fig. 4, the K for the 144 measured sample points ranged from 0.0001 to 0.0674, with a mean value of 0.0243 and a median value of 0.0191. The K obtained by the USLE-K algorithm ranged from 0.0219 to 0.0598, with a mean value of 0.0379 and a median value of 0.0382. The K obtained by the RUSLE2-K algorithm ranged from 0.0131 to 0.0489, with a mean value of 0.0282 and a median value of 0.0291. The K obtained by the EPIC-K algorithm ranged from 0.0190 to 0.0548, with a mean value

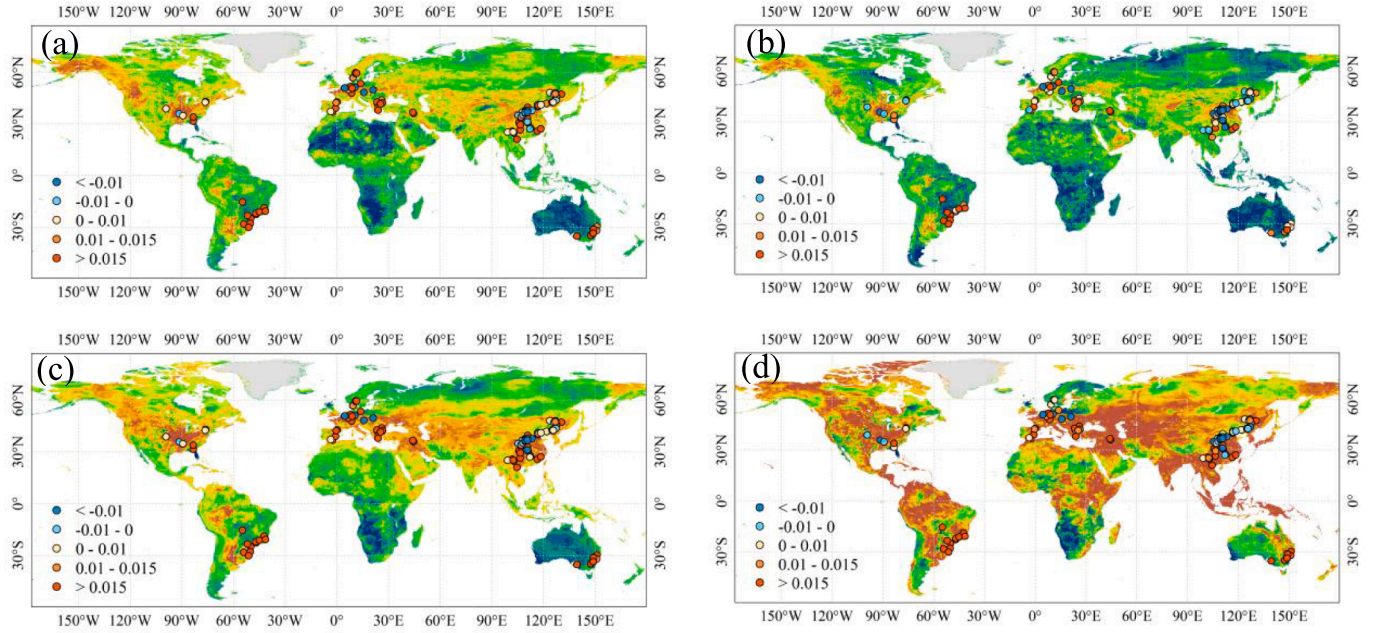


Fig. 5. Difference map of the measured *K* and estimated *K* (a) USLE-K; (b) RUSLE2-K; (c) EPIC-K; (d) Dg-K.

of 0.0397 and a median value of 0.0407. The *K* obtained by the Dg-K algorithm ranged from 0.0112 to 0.0438, with a mean value of 0.0399 and a median value of 0.0399. From the mean value, the result of the RUSLE2-K algorithm is the closest to the measured *K*, followed by the USLE-K algorithm and EPIC-K algorithm. The result of the Dg-K algorithm is quite different from the mean value of the measured *K*.

The difference between the *K* estimated by the four algorithms and the measured *K* was calculated, and the spatial difference between the measured *K* and the estimated *K* (Fig. 5) shows that in terms of spatial distribution, the regions where the *K* obtained by the USLE-K algorithm is closer to the measured *K* (the difference is less than 0.01, accounting for 19.4 %) are mainly located in North America, western Europe and parts of eastern Asia, while South America, most of western Europe, Australia and southern Asia have larger differences between the *K* obtained by the USLE-K algorithm and the measured *K* (the difference is greater than 0.015, accounting for 62.5 %). In contrast, the *K* obtained by the USLE-K algorithm in South America, most of western Europe, Australia and southern Asia differs significantly from the measured values (difference greater than 0.015, accounting for 62.5 %). Compared with the USLE-K algorithm, the *K* obtained by the RUSLE2-K algorithm is closer to the measured values in a larger part (difference less than 0.01, accounting for 43.1 %), the difference is less than 0.01 in most of the regions, and the difference is larger only in some regions in southern Asia, South America and western Europe (difference greater than 0.015, or 36.1 %). The *K* obtained by the EPIC-K algorithm is close to the measured *K* only in a small part of the regions of the Americas and eastern Asia (the difference is less than 0.01, accounting for 18.1 %), while the *K* in the rest of the regions differs significantly from the measured *K* (the difference is greater than 0.015, accounting for 68.1 %). The *K* obtained by the Dg-K algorithm is larger than the measured *K* (the difference is greater than 0.015, accounting for 63.2 %), especially in regions such as southeastern Asia, western Europe and South America, where the difference is larger than the measured *K*, and the regions with smaller differences are only sporadically distributed in North America and Asia (the difference is less than 0.01, accounting for 23.6 %). In summary, the *K* obtained by RUSLE2-K is closer to the measured *K*, followed by the USLE-K and EPIC-K algorithms, and the result of the Dg-K algorithm is quite different from the measured value.

According to the main soil types of 144 measured sample locations, the correlation between the measured *K* of different soil types and the *K* calculated by four algorithms was analyzed (Fig. 6). It can be found that the measured *K* of Luvisols has the highest correlation with the *K* calculated by four algorithms, and the R^2 of *K* obtained by USLE-K, RUSLE2-K and EPIC-K algorithms is above 0.7, and the R^2 of *K* obtained by Dg-K algorithm is 0.453, which has the best correlation with the other three soil types. The R^2 of the measured *K* of Acrisols and Cambisols with the *K* obtained from the USLE-K and RUSLE2-K algorithms can reach more than 0.5, which is a high correlation, but the correlation with the *K* obtained from the EPIC-K and Dg-K algorithms is low, especially the Dg-K algorithm, which has an R^2 of only 0.057 and 0.251. The measured *K* of Ferralsols has the lowest correlation with the *K* obtained from the four algorithms, with R^2 below 0.3. The correlation between the measured *K* of different soil types and the *K* obtained by the four algorithms is different, but in general, the correlation between the measured *K* and the *K* obtained by USLE-K and RUSLE2-K algorithms is greater than that obtained by EPIC-K and Dg-K algorithms.

3.3. Correction of mapping results for the four algorithms based on measured K

Correlation analysis was performed between the measured *K* and the *K* obtained by the four algorithms. A 1:1 graph (Fig. 7) was plotted based on the 144 measured *K* and the *K* obtained by the four algorithms. Most of the *K* values calculated by the USLE-K, RUSLE2-K, EPIC-K and Dg-K algorithms were larger than the measured *K* (86.8 %, 73.6 %, 85.4 % and 79.1 % of overestimation, respectively), and a smaller portion were lower than the measured *K*. The estimation results of all four algorithms on a global scale deviated from the measured *K*. This indicates that the results of these *K* algorithms for estimating *K* need to be corrected as necessary when conditions permit.

Although there are deviations between the estimated and measured *K* of the four algorithms, USLE-K, RUSLE2-K, EPIC-K and Dg-K, all of them show obvious regularity. Further analysis revealed that the following equations exist between the estimated and measured *K* of the four algorithms:

$$K_{USLE-Corrected} = 2.0769K_{USLE} - 0.0543 \quad (25)$$

$$R^2 = 0.611^{**} \quad NSE = 0.61 \quad (p < 0.01)$$

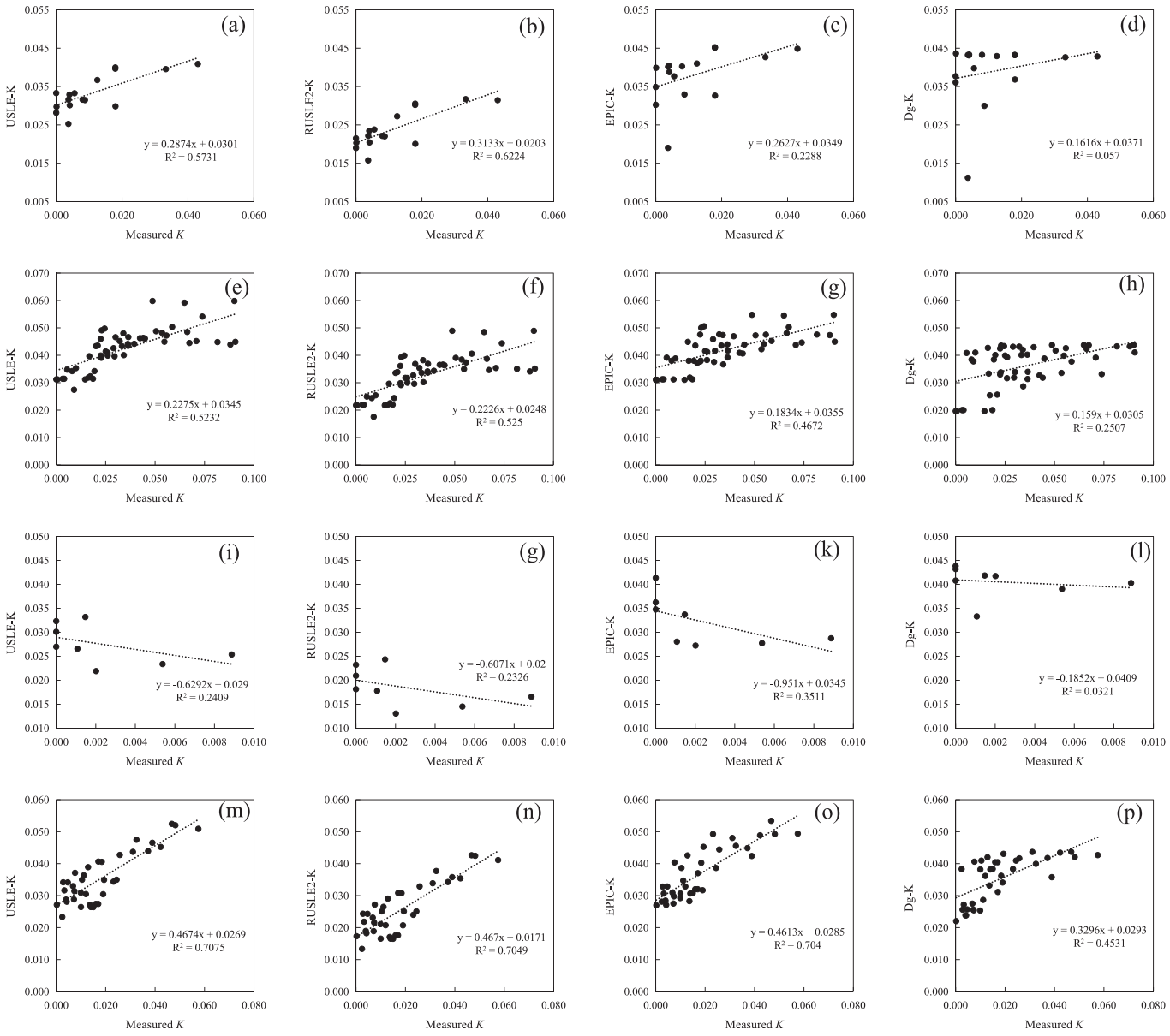


Fig. 6. Correlation analysis of measured *K* and the *K* of four algorithms of different soil types (a) ~ (d) Acrisols;(e) ~ (h) Cambisols; (i) ~ (l) Ferralsols; (m) ~ (p) Luvisols.

$$K_{RUSLE2-Corrected} = 2.1311K_{RUSLE2} - 0.0359 \quad (26)$$

$$R^2 = 0.636^{**} \quad NSE = 0.64 \quad (p < 0.01)$$

$$K_{EPIC-Corrected} = 1.8722K_{EPIC} - 0.05 \quad (27)$$

$$R^2 = 0.414^{**} \quad NSE = 0.41 \quad (p < 0.01)$$

$$K_{Dg-Corrected} = 0.9559K_{Dg} - 0.011 \quad (28)$$

$$R^2 = 0.106 \quad NSE = 0.11$$

The above relationships indicate that the corrected USLE-K algorithm and RUSLE2-K algorithm can meet the model applicability conditions, i.e., $R^2 > 0.5$ and $NSE > 0.4$. The correlations between the *K* obtained by these two corrected algorithms and the measured *K* are roughly comparable. The corrected USLE-K algorithm and RUSLE2-K algorithm are suitable for the estimation of the global *K*. For the EPIC-K algorithm and Dg-K algorithm, the model applicability condition was not met, and the correlation between the *K* obtained by these two algorithms and the measured *K* was low. This result indicates that the results corrected by the above equation can be better used for global soil erosion evaluation, despite the deviations between *K* calculated by the USLE-K and RUSLE2-K algorithms and the measured values (Fig. 8).

4. Discussion

4.1. K And soil science analysis

The statistical results of *K* for different soil types (Table 8, Table 9) indicate that the soil types with low *K* calculated by USLE-K and RUSLE2-K algorithms are Arenosols, Leptosols, Lixisols, Ferralsols and Umbrisols. The reason for this is that Arenosols are sandy and due to the large number of macropores, they can “absorb” a large amount of rain water, which instead of flowing down the surface, soaks into the soil body (Stewart et al., 1975; Vopravil et al., 2007; Dymond et al., 2010). Leptosols are shallow soils on uplands or erosion areas above hard rock or highly calcareous materials that have a high rock fragment content

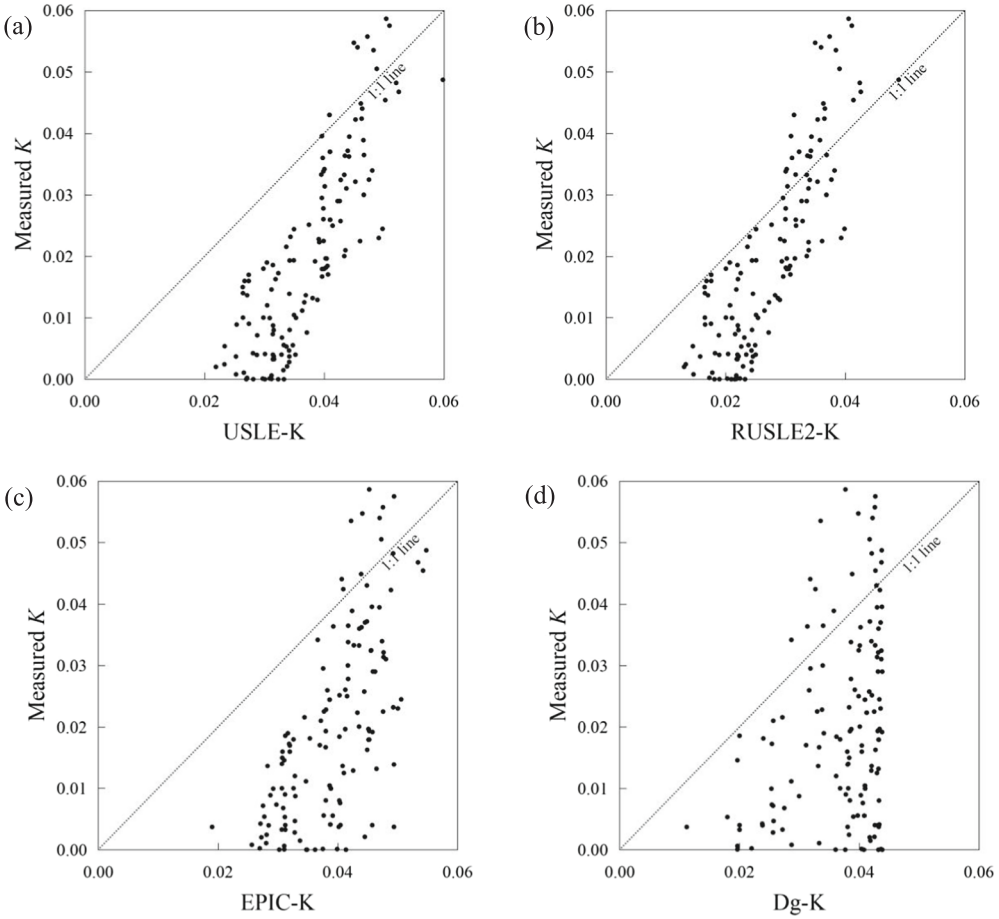


Fig. 7. The 1:1 graph of measured *K* and estimated *K* (a) USLE-K; (b) RUSLE2-K; (c) EPIC-K; (d) Dg-K.

and are more resistant to erosion. Lixisols, Ferralsols and Umbrisols have higher clay content, and therefore have low soil erodibility. The soil types with higher *K* calculated by USLE-K and RUSLE2-K algorithms are Podzols, Cryosols, Histosols, Durisols and Cambisols, because most of these soils have higher silt content and therefore have higher soil erodibility.

The soil types with lower *K* obtained by the EPIC-K algorithm are also Arenosols, Leptosols, Lixisols, Ferralsols and Umbrisols, but the soil types with higher *K* obtained by the EPIC-K algorithm are Calcisols, Fluvisols, Alisols, Chernozems and Solonetz, which are not exactly the same as the soil types with higher *K* obtained by the USLE-K and RUSLE2-K algorithms, probably because the EPIC-K algorithm does not take into account the soil textural class and soil structure class. Soil types with lower *K* obtained by the Dg-K algorithm are Arenosols, Leptosols, Lixisols, Podzols and Regosols, and soil types with higher *K* obtained by the Dg-K algorithm are Phaeozems, Nitisols, Chernozems, Vertisols and Alisols, which differed from the results of the three previous algorithms, probably because the Dg-K algorithm does not take into account the soil organic carbon content, soil texture class and soil structure class.

Soil property data (clay content, silt content, sand content, organic matter content, texture class, and structure class) were used as independent variables in a random forest regression of *K* to obtain the importance of each soil property on *K* (Fig. 9) (The RUSLE2-K algorithm, which has a better relationship between the calculated and measured *K*, is analyzed as an example.), and the MAE was 0.001. The silt content was found to have the greatest effect on *K*, followed by the organic matter content, clay content, structure class and sand content, and the texture class had a relatively small effect on *K*. The reason is that soil particles with particle sizes in the silt size class are more susceptible to erosion, as silty soils are more sensitive to stripping and transport by raindrops and runoff (Bonilla and Johnson, 2012; Ostovari et al., 2019; Vaezi et al., 2010). However, the ease with which soil particles can be separated decreases with decreasing or increasing particle size when the particle size is outside the range of 20–200 μm (Carlos and Odette, 2012; Di and Ferro, 2002). Above this range, soil particles are more difficult to separate and transport due to the mass of the soil particles, e.g., sand, and the erodibility tends to increase with particle size because more energy is required for separating and entraining these particles. In addition, coarse-textured soils typically have higher infiltration rates, resulting in less runoff and thus less erosion. Below this range, soil bonding is stronger, weakening the separation of soil particles, and thus, the clay content has less of an effect on *K* (Carlos and Odette, 2012; Di and Ferro, 2002; Duiker et al., 2001; Romero et al., 2007; Zhang and Mcgrath, 2004). Organic matter acts as a soil binder, flocculating mineral colloids, increasing the stability of soil agglomerates, and decreasing soil detachability; it also increases pore space, increasing infiltration, and decreases splashing when there is not a thin layer of water on the soil, thus decreasing *K* (Mallick et al., 2016; Ostovari et al., 2019; Palmer, 1963). However, the influence of soil physicochemical properties on *K* is more complex, the previous analysis of the relationship between *K* and soil physicochemical properties at the regional scale is weak, and the mechanism of the influence of soil properties on *K* needs to be more deeply and comprehensively studied.

4.2. Comparison with previous studies

A series of previous studies have been performed around global and larger regional *K* mapping. Compared with the global *K* mapping results obtained by these studies, the *K* maps obtained from our calculations (the *K* obtained by the corrected RUSLE2-K algorithm) were similar to

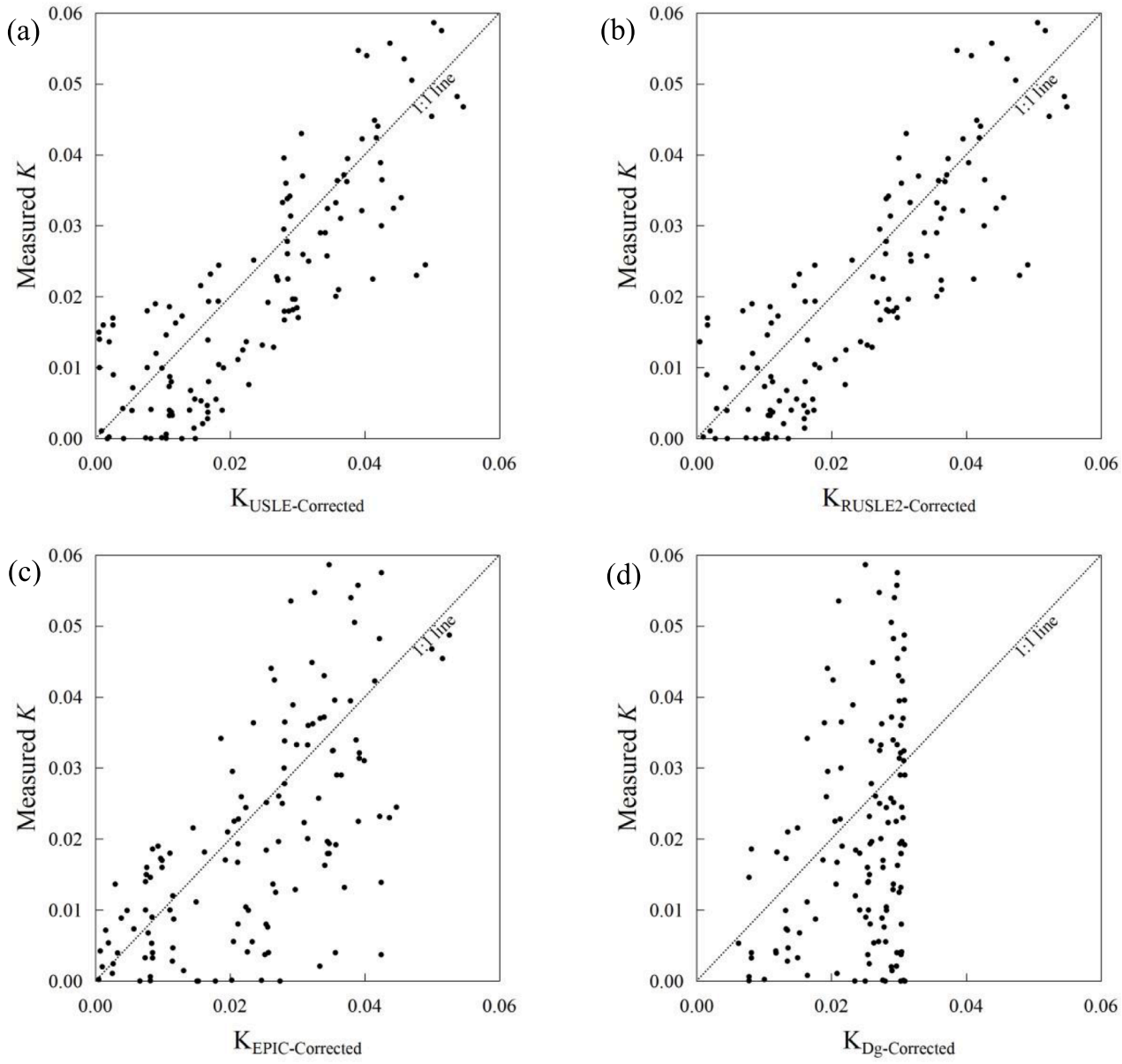


Fig. 8. The 1:1 graph of measured *K* and corrected *K* (a) USLE-K; (b) RUSLE2-K; (c) EPIC-K; (d) Dg-K.

Table 8
Soil types with lower K.

Soil types	USLE-K	Soil types	RUSLE2-K	Soil types	EPIC-K	Soil types	Dg-K
Arenosols	0.014	Arenosols	0.021	Arenosols	0.018	Arenosols	0.016
Leptosols	0.024	Leptosols	0.014	Leptosols	0.021	Leptosols	0.022
Lixisols	0.025	Lixisols	0.015	Ferralsols	0.023	Lixisols	0.026
Ferralsols	0.025	Ferralsols	0.016	Lixisols	0.024	Podzols	0.027
Umbrisols	0.026	Umbrisols	0.014	Umbrisols	0.026	Regosols	0.028

Table 9
Soil types with ligher K.

Soil types	USLE-K	Soil types	RUSLE2-K	Soil types	EPIC-K	Soil types	Dg-K
Podzols	0.063	Podzols	0.054	Calcisols	0.035	Phaeozems	0.041
Cryosols	0.063	Cryosols	0.053	Fluvisols	0.034	Nitisols	0.041
Histosols	0.059	Histosols	0.046	Alisols	0.033	Chernozems	0.041
Durisols	0.056	Durisols	0.046	Chernozems	0.033	Vertisols	0.040
Cambisols	0.050	Cambisols	0.044	Solonetz	0.033	Alisols	0.039

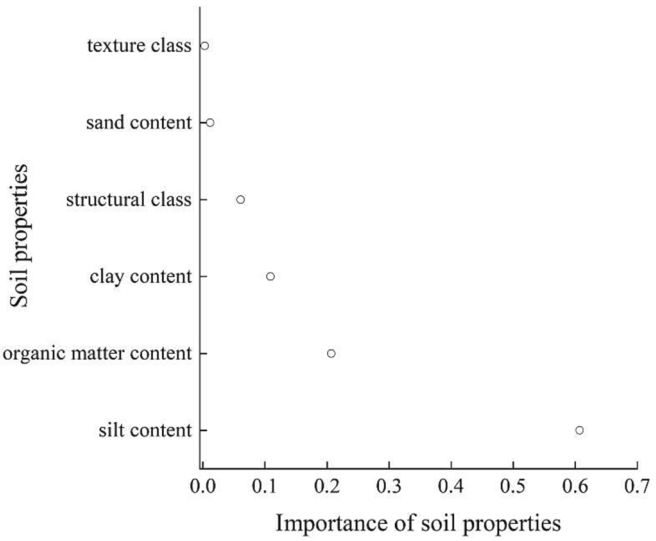


Fig. 9. Importance of soil properties to *K*.

the global *K* maps of Borrelli et al. (2017) (Fig. 10a, 10b) in terms of their spatial pattern distribution, all showing a distribution pattern of higher *K* in Asia, North America and Europe and lower *K* in South America, Africa and Australia. In contrast, comparison with the global *K* map of Nachtergaele et al. (2011) (Fig. 10c, 10d) revealed large spatial differences, such as in southern Africa, northern Asia and Australia. The reason is that Borrelli et al. (2017) and this study both calculate *K* based on soil properties, while Nachtergaele et al. (2011) consider land degradation as a long-term loss of ecosystem function and measure it as a change in net primary productivity (NPP), but a decreasing trend in NPP is also not always considered land degradation, and its results cannot be fully equated with soil degradation.

A comparison of regional mapping results revealed a slightly higher distribution pattern in the central region and a lower distribution pattern in the northern region when compared with the map of Greece by Efthimiou (2020) (Fig. 11a, 11b). Comparison with the map of eastern China by Wang et al. (2016) (Fig. 11c, 11d) shows a distribution pattern of higher values in the east and northeast and lower values in the south. In a comparison with the results of Yang et al. (2018) (Fig. 11e, 11f), both show a higher distribution pattern in the east in New South Wales (NSW) in the map of Australia. Comparison with the map of Brazil by Godoi et al. (2021) (Fig. 11g, 11 h) shows a distribution pattern of higher in the western region of Brazil and lower in the central and eastern regions. In a comparison with the European map by Panagos et al. (2014) (Fig. 11i, 11j), both show a spatial distribution pattern of higher values in central and southern Europe and lower values in northern regions. Overall, the spatial pattern distribution and the range of values of *K* obtained by the corrected RUSLE2-K algorithm in this study are consistent with the existing *K* mapping results.

However, most studies on the applicability of the *K* algorithm have focused on national or regional scale analysis, and due to the large extent of the global scale, it is time-consuming and laborious to measure the *K* values of multiple natural runoff plots in real time, and the dataset for algorithm correction is not large enough. Therefore, there are few studies on the applicability of the *K* algorithm on a global scale, and currently, only Torri et al. (1997) investigated the accuracy of the USLE-K algorithm for *K* prediction using a global *K* dataset extracted from the literature. However, the study did not correct the USLE-K algorithm based on the measured *K* but used soil clay content (*C*), geometric mean particle size (*Dg*), and organic matter content (*OM*) to predict *K* values, and the R^2 was only 0.41. Torri (Torri et al., 1997) also explored an alternative approach to describe *K* based on the upper and lower bounds of the range of *K* variation and the most likely values within that range. However, this approach does not provide information about the observed distribution of *K*.

In conclusion, in this paper, the four *K* algorithms are corrected on the basis of the measured data, and the R^2 between the *K* obtained by the corrected RUSLE2-K algorithm and the measured value can reach 0.636, the NSE can reach 0.64, and the corrected RUSLE2-K algorithm meets the standards required for forecast accuracy and is consistent with the existing regional *K* mapping results in terms of pattern distribution and value range of *K*. The reason is that the RUSLE2-K algorithm calculates *K* by considering the soil erosion process and more comprehensively considers the influence of each factor, which improves the accuracy of the calculation. Therefore, compared with the other three algorithms, the corrected RUSLE2-K algorithm has higher credibility and accuracy,

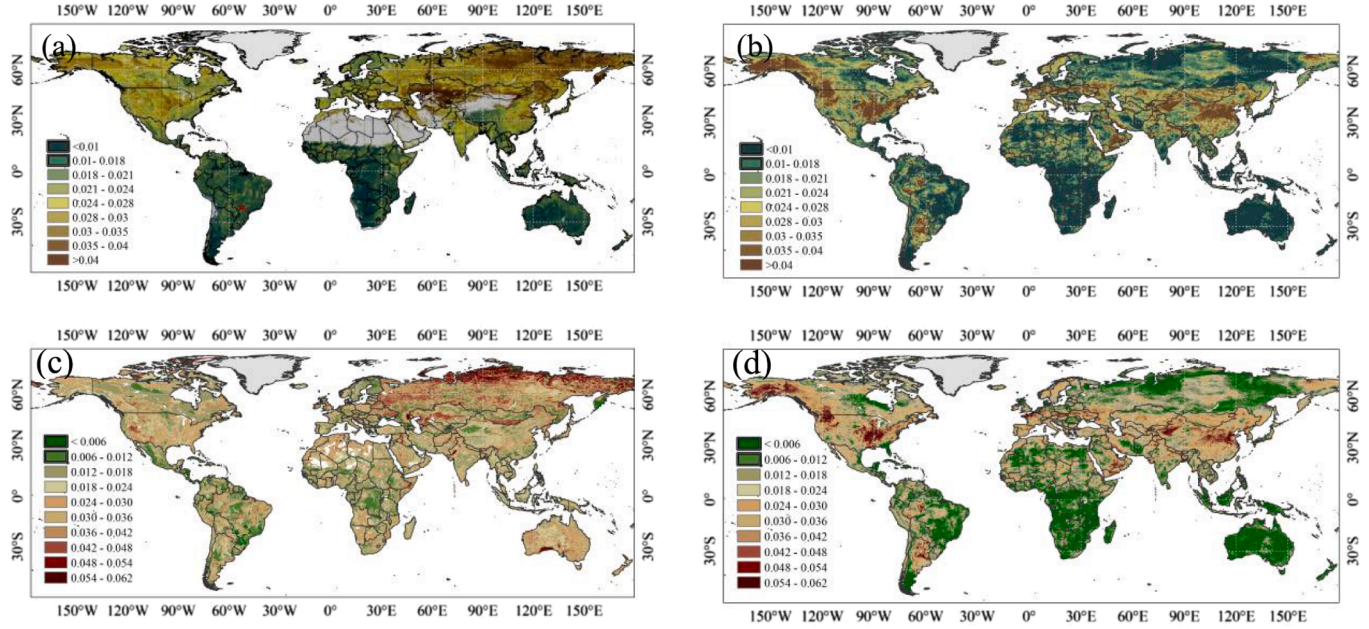


Fig. 10. Comparison of global *K* maps: (a) K_{Borrelli} (Borrelli et al., 2017); (b) $K_{\text{RUSLE2-Corrected}}$; (c) $K_{\text{Nachtergaele}}$ (Nachtergaele et al., 2011); (d) $K_{\text{RUSLE2-Corrected}}$ (Symbol classification is different from (b)).

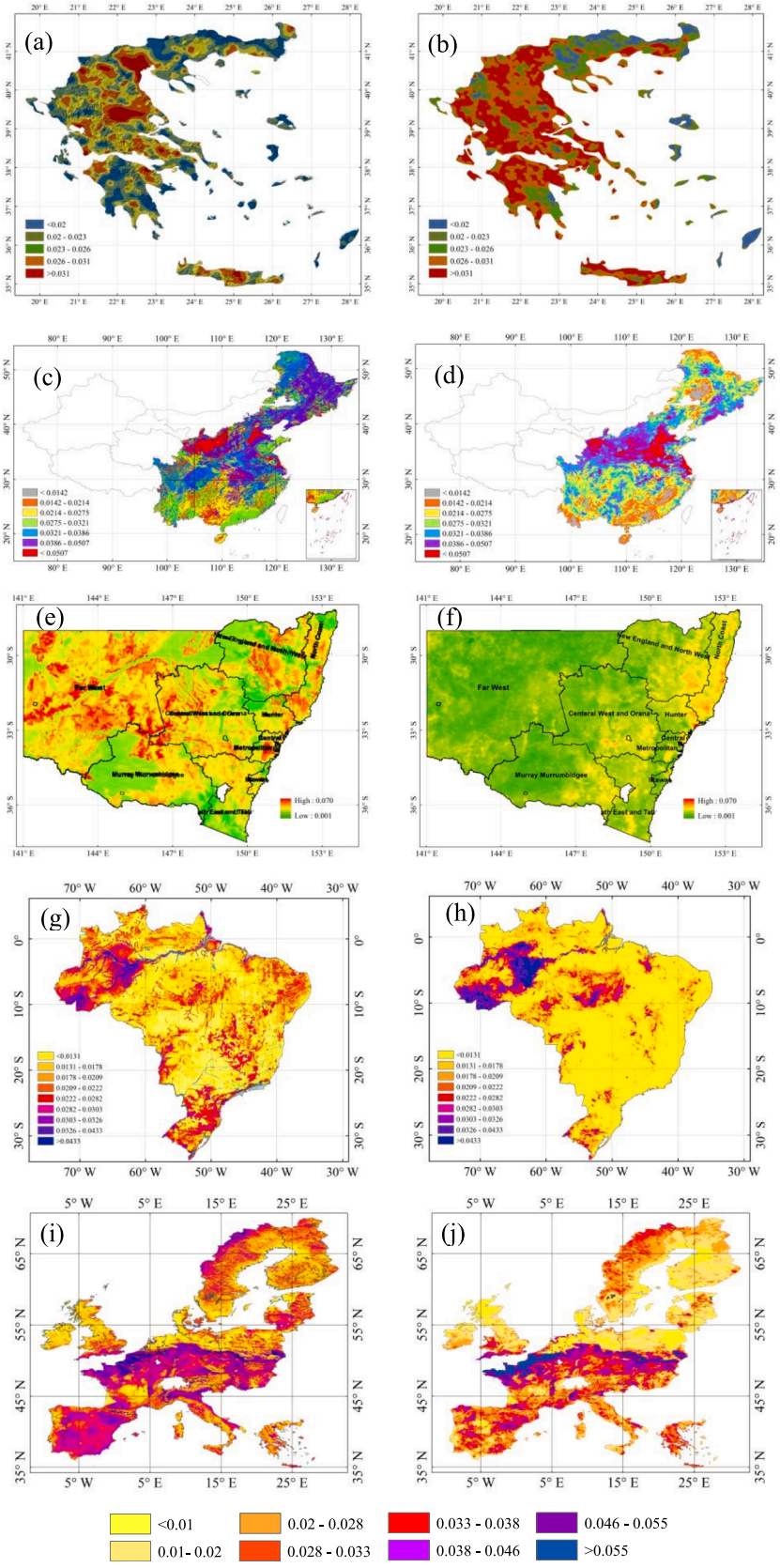


Fig. 11. Comparison of regional *K* mapping results: (a) *K* map of Greece (Efthimiou, 2020); (b) *K* map of Greece-$K_{RUSLE2\text{-}corrected}$; (c) *K* map of eastern China (Yang et al., 2018); (d) *K* map of eastern China-$K_{RUSLE2\text{-}corrected}$; (e) *K* map of New South Wales (Yang et al., 2018); (f) *K* map of New South Wales-$K_{RUSLE2\text{-}corrected}$; (g) *K* map of Brazil (Godoi et al., 2021); (h) *K* map of Brazil-$K_{RUSLE2\text{-}corrected}$; (i) *K* map of Europe (Panagos et al., 2014); (j) *K* map of Europe-$K_{RUSLE2\text{-}corrected}$.

Table A1
Global measured *K* database.

Runoff plot number	Latitude	Longitude	Climate Zones	Country	Region	Precipitation (mm/y)	Land Use Type	Water conservation measures	Slope length (m)	Slope (%)	Area (m^2)	Soil loss rate (t/ha.y)	Measure *K* (Extrapolated) ($t.hm^2.h$)/($hm^2.MJ.mm$)	Measure *K* (literature) ($t.hm^2.h$)/($hm^2.MJ.mm$)	Start year	End year	Publication time	Source
1	−35.43	139.42	BSk	Australia	WELLINGTON	N	Cropland	Cropland	41	8	100	0.03	0.000796793	/	1949	1958	1996	Erskine and Saynor 1996
2	−35.38	139.38	BSk	Australia	WELLINGTON	N	Grassland And Pasture	Cropland	41	8	100	0.1	0.01132717	/	1949	1958	1996	Erskine and Saynor 1996
3	−35.37	139.46	BSk	Australia	WELLINGTON	N	Cropland	Cropland	41	8	100	0.9	0.023452782	/	1949	1958	1996	Erskine and Saynor 1996
4	−35.31	139.39	BSk	Australia	WELLINGTON	N	Cropland	Cropland	41	8	100	1.1	0.026320825	/	1949	1958	1996	Erskine and Saynor 1996
5	−35.09	149.05	Cfb	Australia	CANBERRA	620	Fallow	Fallow	41	8	100	44	0.282440889	/	1979	1981	1996	Erskine and Saynor 1996
6	−35.06	147.55	Cfa	Australia	WAGGA	563	Grassland And Pasture	Lucerne	41	8	100	0.1	0.004238265	/	1947	1976	1996	Erskine and Saynor 1996
7	−35.04	147.6	Cfa	Australia	WAGGA	563	Cropland	Cropland	41	8	100	0.03	0.000226048	/	1947	1976	1996	Erskine and Saynor 1996
8	−35.03	147.47	Cfa	Australia	WAGGA	563	Grassland And Pasture	Pasture	41	8	100	0.04	0.002434687	/	1947	1976	1996	Erskine and Saynor 1996
9	−34.98	147.5	Cfa	Australia	WAGGA	563	Cropland	Cropland	41	8	100	1.6	0.013648554	/	1947	1976	1996	Erskine and Saynor 1996
10	−34.97	147.58	Cfa	Australia	WAGGA	563	Cropland	Cropland	41	8	100	0.9	0.007154621	/	1947	1976	1996	Erskine and Saynor 1996
11	–33.85	148.79	Cfa	Australia	COWRA	N	Grassland And Pasture	Pasture	41	8	100	0.7	0.018817302	/	1943	1954	1996	Erskine and Saynor 1996
12	–33.78	148.92	Cfb	Australia	COWRA	N	Grassland And Pasture	Lucerne	41	8	100	0.8	0.025440597	/	1943	1954	1996	Erskine and Saynor 1996
13	–33.77	148.72	Cfa	Australia	COWRA	N	Cropland	Cropland	41	8	100	0.7	0.003961253	/	1949	1954	1996	Erskine and Saynor 1996
14	–33.74	148.81	Cfa	Australia	COWRA	N	Cropland	Cropland	41	8	100	1.7	0.009936057	/	1943	1975	1996	Erskine and Saynor 1996
15	–33.64	148.73	Cfa	Australia	COWRA	N	Cropland	Cropland	41	8	100	0.6	0.00327937	/	1943	1975	1996	Erskine and Saynor 1996
16	–33.64	148.89	Cfa	Australia	COWRA	N	Grassland And Pasture	Pasture	41	8	100	0.2	0.00734516	/	1943	1975	1996	Erskine and Saynor 1996
17	–33.61	148.79	Cfa	Australia	COWRA	N	Cropland	Cropland	41	8	100	1.2	0.006784021	/	1955	1975	1996	Erskine and Saynor 1996

(*continued on next page*)

Table A1 (*continued*)

Runoff plot number	Latitude	Longitude	Climate Zones	Country	Region	Precipitation (mm/y)	Land Use Type	Water conservation measures	Slope length (m)	Slope (%)	Area (m^2)	Soil loss rate (t/ha.y)	Measure *K* (Extrapolated) (t.hm^2.h)/(hm^2.MJ.mm)	Measure *K* (literature) (t.hm^2.h)/(hm^2.MJ.mm)	Start year	End year	Publication time	Source
18	–32.87	−68.87	BWK	Argentina	CAP3	N	Shrubland	/	10	14.6	30	0.1876	0.002284753	/	1983	1994	2001	Braud et al. 2001
19	–32.87	−68.87	BWK	Argentina	CAP2	N	Shrubland	/	10	16.7	30	0.3791	0.003924214	/	1983	1994	2001	Braud et al. 2001
20	–32.87	−68.87	BWK	Argentina	CAP1	N	Fallow	/	10	19.6	30	1.4517	0.005107351	/	1983	1994	2001	Braud et al. 2001
21	−30.1	−51.32	Cfa	Brazil	Porto Alegre	1500	Forest	/	12	20	48	4.6E-11	1.58983E-12	/	2006	2007	2013	Oliveira et al. (2013a)
22	−30.1	−51.32	Cfa	Brazil	Porto Alegre	1500	Forest	/	12	45	48	1.6E-10	2.29746E-12	/	2006	2007	2013	Oliveira et al. (2013a)
23	−30.1	−51.32	Cfa	Brazil	Porto Alegre	1500	Forest	/	12	20	48	8E-11	2.76492E-12	/	2006	2007	2013	Oliveira et al. (2013a)
24	−30.1	−51.32	Cfa	Brazil	Porto Alegre	1500	Forest	/	12	8	48	3E-11	3.64991E-12	/	2006	2007	2013	Oliveira et al. (2013a)
25	−30.1	−51.32	Cfa	Brazil	Porto Alegre	1500	Fallow	/	12	8	48	2.228E-09	5.21069E-12	/	2006	2007	2013	Oliveira et al. (2013a)
26	−30.1	−51.32	Cfa	Brazil	Porto Alegre	1500	Fallow	/	12	20	48	1.0243E-08	6.80516E-12	/	2006	2007	2013	Oliveira et al. (2013a)
27	−30.1	−51.32	Cfa	Brazil	Porto Alegre	1500	Fallow	/	12	20	48	1.4637E-08	9.72441E-12	/	2006	2007	2013	Oliveira et al. (2013a)
28	−30.1	−51.32	Cfa	Brazil	Porto Alegre	1500	Fallow	/	12	45	48	6.3795E-08	1.7609E-11	/	2006	2007	2013	Oliveira et al. (2013a)
29	−29.98	−51.3	Cfa	Brazil	Terra Dura	1542	Forest	/	12	8.79	48	3.7E-11	4.45537E-12	/	2006	2010	2011	Oliveira (2011)
30	−29.98	−51.3	Cfa	Brazil	Terra Dura	1542	Fallow	/	12	8.79	48	7.01E-09	1.56498E-11	/	2006	2010	2011	Oliveira (2011)
…	…	…	…	…	…	…	…	…	…	…	…	…	…	…	…	…	…	…
1657	–33.8	148.72		Australian	Cowra	/	Fallow	/	42	4	/	/	/	0.044	2000	2012	2017	Yang2017
1658	–33.81	148.7		Australian	Cowra	/	Fallow	/	42	11	/	/	/	0.031	2000	2012	2017	Yang2017
1659	–33.81	148.7		Australian	Cowra	/	Fallow	/	42	4	/	/	/	0.037	2005	2010	2017	Yang2017
1660	−31.03	150.27		Australian	Gunnedah	/	Fallow	/	42	8	/	/	/	0.046	2005	2010	2017	Yang2017

can be used for global *K* mapping and has a certain reference value for the prediction and correction of global *K*.

4.3. Influence factors and uncertainty in K

Theoretically, *K* is only related to soil properties. However, there are also some studies on the relationship between *K* and environmental conditions, such as soil-forming parent material (Gao et al., 2022), elevation (Zhu et al., 2021), slope position (Chen et al., 2022), vegetation communities (Ostovari et al., 2018; Walia and Dick, 2018), stage of soil profile truncation (Radziuk and Świtoniak, 2021) and land-use change (Wang et al., 2019), among others. In reality, these factors mainly affect *K* by influencing the soil properties and thus *K* (Wischmeier and Smith, 1978; Renard and Dabney, 2013; Sharpley & Williams, 1989). Therefore, the study of the factors influencing *K* is pending a deeper explanation and analysis based on the effects of these factors on soil properties.

Soil erodibility is inherently complex due to the complexity of the erosion process, leading to some uncertainty in the calculation of soil erodibility (Römkens et al., 1997). For example, the length of observation time series for runoff plots, rainfall simulators for rainfall types of simulated rainfall experiments, different soil loss equations and different measurement of soil property data lead to the possibility of some errors in the extracted *K* as well (Römkens, 1985). Errors in sampling, measurement, and modeling will result in nonnegligible uncertainty (Buttafuoco et al., 2012), and understanding how uncertainty in model parameters and data propagates through the model is significant for accurate prediction of *K*.

Most of the available uncertainty analyses have been carried out in relatively small areas. For example, Madenoglu et al. (2020) assessed the spatial uncertainty of *K* due to differences in soil types using kriging and direct sequential simulation (DSIM) in the Asartepe dam basin, Turkey, and Zhang et al. (2009) evaluated the uncertainty of different *K* prediction models based on the observed values of *K* in seven typical soils in subtropical China and selected mathematical statistical terms such as the mean absolute error (MAE) as indicators. Castrignanò et al. (2008) evaluated the spatial uncertainty of the *K* model due to uncertainties in the input parameters (texture and organic matter) using Monte Carlo analysis based on 152 samples from the east-central part of Sardinia (Italy). However, there are fewer studies of soil erosion models predicting uncertainty in *K* values on a large regional and global scale, and some mathematical methods (e.g., Borrelli et al. (2017)) used for uncertainty evaluation of the soil loss rate (*A*), which is based on various factors, may be a viable approach for uncertainty analysis. However, due to the lack of sufficient global measured data, the uncertainty analysis part of the four algorithms cannot be simulated or interpolated by machine learning and sequential Gaussian simulation methods for the time being, and uncertainty analysis, especially accuracy evaluation, has only roughly clarified the idea at present. Consequently, there are still difficulties. In the future, the measured *K* values in newly published literature can be tracked, the sample size of measured *K* values can be expanded, and the uncertainties of the four algorithms can be systematically analyzed.

4.4. Limitations of the study and future work

In this study, due to the limited data of soil erosion test observations (plots) in the literature that also include land use type, biological/engineering measures, plot length, plot slope, and water erosion loss rate, only data from 1260 runoff plots and 144 actual sample points were collected in this study, which is a small sample size. On the other hand, the rainfall and runoff factors (*R*), slope length and slope factors (*L*, *S*), cover and management measures factors (*C*), and soil and water conservation measures factors (*P*) used to derive the measured *K* in this study are mainly calculated or assigned based on rainfall data, vegetation cover data, land use types, and slope and tillage types extracted from the literature, which vary over time and space, and the soil erodibility values of specific soils may vary dynamically in space, which leads to the fact that the measured *K* derived in this study may also have some errors (Wang et al., 2001). Therefore, further research is needed to improve the accuracy of the measured data and its calculation.

From the previous analysis, it can be seen that the RUSLE2-K algorithm is the closest to the measured *K*, mainly because the algorithm considers a fuller range of soil properties. The RUSLE2-K algorithm has some minor improvements on the basis of the USLE-K algorithm, such as the influence of the soil structure class, in the USLE-K algorithm $k_{s\ \mathrm{USLE}} = 3.25(\mathrm{s}\text{-}2)$, and in the RUSLE2-K algorithm $k_{s\ \mathrm{RUSLE2}} = 3.25(2\text{-}\mathrm{s})$; this improvement is theoretically correct but the analysis of the specific influence mechanism is still difficult. Meanwhile, it has been demonstrated that *K* is also affected by rock fragments in the topsoil profile and rock fragments on the soil surface (Zhu and Shao, 2008; Li et al., 2007; Yang et al., 2023), etc. However, since the effect of rock fragment are usually not taken into account in plot tests, or rock fragment cover/content information is not documented in the literature, the correction of *K* algorithm did not consider the effect of rock fragment and other influencing factors in this study. With the accumulation of more measured data, it may be possible to try to realize high-precision mapping of *K* based on deep learning methods, analyze the applicability of the four algorithms to different regions, analyze the reasons for their applicability, and improve them gradually, so as to obtain a *K* algorithm that is more suitable for the whole world or a revised algorithm.

5. Conclusion

This study uses ISRIC 2.0 soil property data to estimate the global soil erodibility factor (*K*) based on four algorithms, USLE-K, RUSLE2-K, EPIC-K and Dg-K, and compares the spatial patterns of different global *K* calculation results obtained by the four algorithms, extrapolates measured *K* data using literature backtracking methods and search tool methods, establishes a database of measured *K*, analyzes the results of the above four algorithms for applicability, and corrects the above four algorithms based on the database of measured *K*. The main conclusions of this study are as follows:

(1) The spatial patterns of global *K* obtained by the four algorithms of USLE-K, RUSLE2-K, EPIC-K and Dg-K are similar, showing a distribution pattern of lower *K* in South America, northern and southern Africa, and Europe and higher *K* in southern North America, western Europe, northern and central Asia, and the distribution characteristics of *K* show a certain regularity with the distribution of soil properties.

(2) The estimation results of different *K* estimation methods vary slightly globally, and when compared with the established database of measured *K*, the RUSLE2-K algorithm is the closest to the measured *K*, followed by the USLE-K algorithm and EPIC-K algorithm, and the Dg-K algorithm differs significantly from the measured *K* in terms of the mean values and the spatial difference maps between the measured *K* and the estimated *K*.

(3) Globally, the correlation between the *K* obtained by EPIC-K algorithm and Dg-K algorithm and the measured *K* was low, which failed to meet the model applicability condition. Although there were deviations between the *K* calculated by the RUSLE2-K and USLE-K algorithms and the measured *K*, the corrected results could meet the model applicability conditions and are generally consistent with the existing *K* mapping results. Therefore, they are suitable for the estimation of the global *K* and potentially represent a good opportunity for global soil erosion evaluation.

CRediT authorship contribution statement

Miaomiao Yang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Data curation, Conceptualization. **Qinke Yang:** Writing – review & editing, Supervision, Software, Project administration, Methodology, Conceptualization. **Keli**

Zhang: Writing – review & editing, Software, Conceptualization. **Guowei Pang:** Writing – review & editing, Methodology. **Chenlu Huang:** Writing – review & editing, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Extracts of relevant information from literature on *K* are recorded in the global measured *K* database. Since the database contains data from 1260 runoff plots, only a portion of the global soil erodibility factor (*K*) is presented in Table A1.

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